A Comparative Evaluation of Prompt Engineering and Retrieval-Augmented Generation for AI-Based Customer Service in Philippine Dental Clinics

by

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RECOMMENDATION SHEET

This is to certify that I have supervised the preparation of and read the thesis proposal prepared by **Irvan F. Castro, Alexis Arvie S. Manuel**, and **Gabriel Matthew R. Roda** entitled ***A Comparative Evaluation of Prompt Engineering and Retrieval-Augmented Generation for AI-Based Customer Service in Philippine Dental Clinics*** and that the said paper has been submitted for final examination by the Oral Examination Committee.

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ABSTRACT

Large language models (LLMs) have significantly advanced conversational AI, allowing systems to generate context-aware responses that are suited to specific users. However, challenges remain in deploying these models in domain-specific contexts, including factual inaccuracies, limited adaptability, and inefficiencies. These challenges are particularly acute in specialized, high-reliability settings such as healthcare customer service. This research proposes a study to evaluate two primary optimization techniques—Prompt Engineering and Retrieval-Augmented Generation (RAG)—to address these limitations for AI-powered customer service agents in Philippine dental clinics. Focusing on appointment management and patient interaction, the study will conduct a controlled, head-to-head evaluation of both methods. Using realistic datasets comprising clinic FAQs, service information, and simulated patient dialogues, the research will assess their effectiveness along key metrics, including accuracy, robustness to noisy inputs (e.g., Taglish, typos), scalability with expanding knowledge bases, and cost efficiency. This study aims to provide a technical foundation and empirical evidence for developing reliable, domain-adaptive, and knowledge-enhanced conversational AI systems for practical deployment in healthcare.

Keywords: Retrieval-Augmented Generation (RAG), Prompt Engineering, Large Language Models (LLMs), Conversational AI, Healthcare Customer Service

Chapter I

INTRODUCTION

Background of the Study

Artificial intelligence (AI) has become a central area of research in computer science, with natural language processing (NLP) and conversational systems emerging as prominent application domains. The development of large language models (LLMs) based on transformer architectures has significantly advanced the ability of machines to generate context-aware, human-like responses. As a result, LLMs have been increasingly integrated into tasks such as information retrieval, customer service automation, and administrative workflow support (Lewis et al., 2020; Fan et al., 2024; Singh & Namin, 2025). These capabilities position LLM-based conversational agents as scalable tools for managing high-volume, text-based interactions between users and service providers.

Despite these advancements, the deployment of LLM-driven conversational agents in domain-specific environments remains challenging. In healthcare-related customer service contexts, including dental clinics, conversational systems must handle structured workflows such as appointment booking, rescheduling, and follow-ups, while maintaining factual accuracy and contextual consistency. Many existing systems continue to rely on rule-based chatbot architectures that depend on predefined keywords and decision trees (Shin et al., 2024; Singh et al., 2022). While such systems offer predictable behavior, they exhibit limited adaptability, struggle with linguistic variation, and often fail to maintain relevance across multi-turn conversations. These limitations are particularly evident in multilingual environments such as the Philippines, where users frequently employ code-switching (e.g., Taglish), informal expressions, and misspellings, leading to reduced user satisfaction (Shin et al., 2024). The dental clinic domain presents additional domain-specific considerations that make it a meaningful environment for evaluating conversational AI optimization techniques. Unlike general service businesses such as salons or spas, dental clinics operate within a healthcare context where informational accuracy is critical. Patient inquiries frequently involve treatment explanations, procedural preparation, pricing variations, and post-treatment guidance, all of which require precise and reliable responses. Inaccurate or hallucinated information in such contexts may lead to patient misunderstanding or reduced trust in clinical services. These requirements place stronger demands on conversational systems to maintain factual grounding and contextual consistency. Consequently, the dental clinic setting provides a structured yet knowledge-intensive customer service environment that is well suited for evaluating how Prompt Engineering and Retrieval-Augmented Generation (RAG) handle domain-specific knowledge retrieval, hallucination mitigation, and conversational workflow management.

LLM-based agents address several limitations of rule-based systems by offering flexible language understanding and generative capabilities. However, their deployment introduces new system-level challenges, including hallucinated or factually incorrect responses, degradation of context over extended interactions, increased latency, and rising computational costs as contextual information scales (Holst & Iala, 2024; Lewis et al., 2020). These issues are especially problematic in appointment-oriented customer service settings, where accuracy, consistency, and responsiveness are critical for maintaining user trust.

Statement of the Problem

The effective deployment of LLM-based conversational agents in appointment-focused customer service environments presents a core computer science problem: how to achieve accurate, robust, and contextually consistent responses while maintaining scalability and computational efficiency under real-world interaction constraints. In Philippine dental clinics, conversational agents must operate under several practical conditions, including multilingual and noisy user inputs (such as Taglish and typographical errors), multi-turn conversational workflows, and continuously expanding domain knowledge bases.

Two widely adopted strategies for addressing these challenges are Prompt Engineering and Retrieval-Augmented Generation (RAG). Prompt Engineering attempts to guide model behavior through carefully structured instructions and contextual injection, while RAG enhances response generation by retrieving relevant information from an external knowledge base and grounding outputs in retrieved documents. Each approach manipulates how contextual information is supplied to the language model and introduces distinct trade-offs across key system parameters, including response accuracy, hallucination rate, robustness to linguistic variation, multi-turn context retention, response latency, token usage, and scalability as knowledge bases grow.

Adjustments to these parameters can improve certain aspects of system performance while negatively impacting others. For example, injecting large amounts of contextual information through prompts may improve factual accuracy but increase token cost and latency, whereas retrieval-based methods may reduce hallucinations at the expense of added computational overhead. Despite the practical importance of these trade-offs, there is limited empirical evidence comparing how Prompt Engineering and RAG perform relative to one another under controlled, real-world conditions.

As a result, developers and practitioners lack a clear basis for selecting an appropriate optimization strategy for LLM-driven conversational agents deployed in localized, multilingual customer service environments. This uncertainty motivates the need for a systematic and comparative evaluation of Prompt Engineering and Retrieval-Augmented Generation, focusing on how each approach balances accuracy, robustness, scalability, and computational efficiency in appointment-oriented conversational workflows.

Research Questions

This study aims to examine the comparative performance of Prompt Engineering and Retrieval-Augmented Generation (RAG) as contextual optimization techniques in a multilingual dental clinic customer-service setting. Specifically, this study seeks to answer the following research questions:

Main Research Question

How do Prompt Engineering and Retrieval-Augmented Generation (RAG) compare in terms of response accuracy, hallucination rate, scalability, computational efficiency, and contextual consistency in a multilingual dental clinic customer-service chatbot system?

Sub-Questions

1. How does increasing knowledge-base size affect the response accuracy and system latency of Prompt Engineering and RAG?
2. How does the presence of noisy multilingual inputs (e.g., typographical errors, informal Taglish expressions, incomplete queries) affect the robustness and reliability of each optimization technique?
3. How does increasing multi-turn conversational depth influence contextual consistency and response stability for both methods?
4. How do Prompt Engineering and RAG differ in terms of token usage and computational cost during runtime?
5. Is there a statistically significant difference between the performance metrics of the two techniques across defined experimental conditions?

Research Objectives

In response to the statement of the problem, the general objective of this study is to evaluate and compare the effectiveness of Prompt Engineering and Retrieval-Augmented Generation (RAG) in optimizing AI-powered customer service agents for Philippine dental clinics within a controlled, chat-based web environment. Specifically, the study aims to examine how the two approaches differ in their ability to generate accurate, factual, and contextually appropriate responses under varied input conditions, including multilingual usage, typographical errors, and informal language. The study further seeks to assess how effectively each approach maintains coherence and relevance across multi-turn conversations related to appointment booking, follow-ups, and rescheduling.

In addition to response quality, the study aims to compare the computational performance of both approaches by measuring response latency, token consumption, and scalability as the size and complexity of the clinic knowledge base increase. To achieve these objectives, the study will evaluate generated responses against a curated ground-truth knowledge base, systematically identify hallucinations and error types, test robustness using noisy and ambiguous inputs, assess context handling in simulated multi-turn workflows, and analyze efficiency and cost-related metrics.

Hypotheses

Given the comparative and experimental nature of this study, the following hypotheses are formulated to guide statistical analysis and determine whether measurable performance differences exist between Prompt Engineering and Retrieval-Augmented Generation (RAG) under controlled conditions.

H1: Retrieval-Augmented Generation (RAG) achieves significantly higher factual accuracy than Prompt Engineering as the knowledge base increases in size.

H2: Prompt Engineering produces lower response latency compared to Retrieval-Augmented Generation (RAG) when operating under small knowledge base conditions.

H3: Retrieval-Augmented Generation (RAG) yields a significantly lower hallucination rate than Prompt Engineering when processing noisy, multilingual, or ambiguous user inputs.

H4: Knowledge base size has a statistically significant effect on system performance metrics, including accuracy, latency, and token consumption, for both optimization techniques.

These hypotheses reflect the underlying trade-offs identified in the problem statement—namely, the balance between contextual grounding, scalability, and computational efficiency. The statistical testing of these hypotheses enables the study to move beyond descriptive comparison and determine whether observed performance differences are significant rather than incidental variations.

Significance and Scope of the Study

This study is significant because it addresses key limitations of LLM-based conversational agents in domain-specific, multilingual customer service environments. By providing a controlled, empirical comparison of Prompt Engineering and RAG, the research contributes evidence-based insights into the trade-offs between accuracy, robustness, scalability, and cost efficiency. These findings advance computer science research in NLP system design, domain adaptation of LLMs, hallucination mitigation, and cost-aware retrieval strategies. For healthcare service providers such as dental clinics, the findings may provide guidance on selecting appropriate conversational AI architectures for managing patient inquiries while minimizing the risk of misinformation in treatment-related or service-related communication.

The study focuses on customer service automation for Philippine dental clinics, evaluated within a controlled chat-based web environment designed to simulate real-world patient–clinic interactions. Its scope is limited to handling informational queries and appointment-related workflows, including booking, rescheduling, and cancellation. The underlying language model will be held constant to isolate the effects of the optimization strategies. The study does not address voice-based interfaces, clinical diagnosis, or integration with production booking or health record systems.

Limitations of the Study

Despite the structured experimental design employed in this study, several limitations must be acknowledged. First, the study utilizes a single large language model backend, GPT-4o mini, to ensure experimental control. While this allows for a fair comparison between Prompt Engineering and Retrieval-Augmented Generation (RAG), the results may not fully generalize to other LLM architectures or model sizes that may exhibit different response behaviors.

Second, the evaluation is conducted within a controlled experimental environment using simulated conversational datasets rather than live clinical deployments. Although the scenarios were designed to reflect realistic patient–clinic interactions, actual user behavior, system load, and operational constraints in real dental clinics may introduce additional variables not captured in the simulation.

Third, the knowledge base and conversational scenarios were constructed from curated public information and predefined interaction flows. As a result, the system was not evaluated using real patient data or live booking systems. Consequently, the study does not assess the performance of the conversational agents when integrated with real-time appointment scheduling systems or electronic health record infrastructures.

Fourth, the experimental configuration fixes several system parameters, including the embedding model (text-embedding-3-large) and retrieval depth ($k = 5$) for the RAG configuration. While these settings provide consistency across experimental runs, alternative embedding models or retrieval configurations may produce different performance outcomes.

Additionally, the study focuses exclusively on text-based chat interactions and does not evaluate voice-based conversational interfaces or multimodal input systems. Finally, the linguistic evaluation primarily considers English, Tagalog, and Taglish conversational inputs common in Philippine dental clinic contexts, which may limit the applicability of the findings to other linguistic environments.

These limitations should be considered when interpreting the results of the study and when applying its findings to real-world deployments of conversational AI systems.

Chapter II

REVIEW OF RELATED LITERATURE

This review synthesizes foundational concepts and recent studies relevant to the development of domain-specific conversational agents. It first establishes the theoretical background for the core components of this research, including AI-powered customer service agents, Large Language Models (LLMs), Prompt Engineering, and Retrieval-Augmented Generation (RAG). It then provides a synthesized analysis of related studies to identify current trends, persistent gaps, and the technical foundation upon which this proposal is built.

Theoretical Background and Related Concepts

AI-Powered Customer Service Agents

AI-powered customer service agents are conversational systems designed to automate user interactions. They range from rule-based chatbots to advanced LLM-driven agents capable of understanding context and generating natural language responses. Modern systems can handle multiple tasks simultaneously, such as answering inquiries, scheduling appointments, and providing recommendations. The effectiveness of these agents relies on robust NLP pipelines, knowledge integration, and context-awareness. Key technical challenges include response relevance, scalability, handling ambiguous queries, and maintaining system efficiency under concurrent requests.

Large Language Models (LLMs)

Large language models (LLMs), including GPT, BERT, and other transformer-based architectures, form the backbone of state-of-the-art conversational agents. These models are pre-trained on massive text corpora using self-supervised learning and can be fine-tuned for domain-specific tasks. Transformers leverage attention mechanisms to capture long-range dependencies in text, enabling coherent and context-sensitive outputs. Key technical considerations for LLM deployment include model size and inference latency, domain adaptation, and ensuring response consistency and factual accuracy.

Prompt Engineering

Prompt engineering is the process of designing and refining input instructions to guide LLM outputs toward desired behavior. It plays a critical role in domain-specific applications by improving response accuracy and controlling model behavior. Techniques include zero-shot and few-shot prompting, chain-of-thought prompting to improve logical consistency, and instruction tuning to align with specific tasks, user intents, or domain constraints.

Retrieval-Augmented Generation (RAG)

RAG integrates LLMs with external knowledge retrieval mechanisms. Instead of relying solely on pre-trained knowledge, the system retrieves relevant documents or data from structured or unstructured sources and conditions the generation process on this information. Key components include a knowledge base, embedding and vector databases, and a retriever–generator architecture. RAG enhances domain specificity, factual correctness, and response reliability, addressing common limitations of pure generative models (Lewis et al., 2020).

Knowledge Bases and Data Management

A robust knowledge base is essential for RAG systems. Key aspects include data curation to ensure accuracy and relevance, efficient indexing and retrieval through optimized embeddings, and update mechanisms to reflect dynamic information such as appointment schedules or clinic policies.

NLP Evaluation Metrics

Evaluating AI-powered conversational agents requires both automatic metrics (e.g., BLEU, ROUGE, perplexity) and human-centered assessments. In customer service domains, human-centered metrics such as accuracy, informativeness, and user satisfaction are critical. Task-oriented metrics, like the success rate in correctly booking an appointment, are also essential.

Human–AI Interaction Principles

Technically optimized systems must also account for usability and user trust. Core principles include explainability of AI outputs, robust error handling, and cultural and linguistic adaptation. For the Philippine context, this includes multilingual support for English and Tagalog and context-aware communication styles.

Synthesis of Related Studies

The reviewed literature reveals several consistent trends in the development of retrieval-augmented generation (RAG) and prompt-engineered large language models (LLMs) for specialized domains. A primary observation is that integrating retrieval mechanisms with generative models substantially improves factual accuracy and relevance compared to generative-only approaches. The foundational work by Lewis et al. (2020) introduced RAG as a method to combine parametric and non-parametric memory, demonstrating state-of-the-art results on open-domain question answering tasks. Subsequent research has confirmed that this architecture effectively mitigates hallucinations and allows knowledge to be updated dynamically, addressing a critical limitation of static LLMs (Gupta et al., 2024; Fan et al., 2024).

In specialized domains such as healthcare, this grounding mechanism becomes particularly important. Studies on domain-specific chatbots show that RAG-based systems significantly enhance reliability and factual consistency. For example, Arian et al. (2024) proposed a prototype “Dental Loop Chatbot” using RAG and fine-tuning, which improved accuracy in responding to dental-related queries. Similarly, Steybe et al. (2025) found that a RAG-powered chatbot designed to answer clinical questions related to medication-related osteonecrosis of the jaw (MRONJ) achieved higher scores in informativeness and scientific explanation compared with baseline LLM responses. This trend is further supported by domain-adaptation research such as Siriwardhana et al. (2023), which demonstrated that jointly optimizing the retriever and generator can improve performance when applying RAG to specialized knowledge corpora.

Alongside retrieval-based approaches, prompt engineering has been identified as a key method for guiding LLM behavior and ensuring task alignment. Wang et al. (2025) investigated prompt engineering in an AI-driven healthcare system and found that carefully structured prompts significantly increased response accuracy, improving performance from 80.1% to 99.6%. However, the study also reported increased response latency and slightly lower perceived user satisfaction, highlighting trade-offs between accuracy and system responsiveness. Other studies, including Waaler et al. (2025) and Patil et al. (2024), further demonstrate that structured prompt design can enhance response reliability and adherence to domain-specific guidelines in healthcare and mental health chatbot applications.

Some recent studies have also explored broader comparisons of LLM optimization strategies. For instance, comparative analyses examining prompt engineering, retrieval-augmented generation, and fine-tuning highlight trade-offs between implementation complexity, data requirements, and factual grounding. These studies generally conclude that prompt engineering provides a lightweight approach for improving model behavior, while retrieval-based augmentation offers stronger factual grounding for knowledge-intensive tasks. However, such comparisons are typically conducted at the model or task level and do not evaluate how these techniques perform within conversational systems or operational service environments.

Despite these advances, several limitations remain in the existing literature. Many studies remain at the prototype stage, with evaluations conducted using small curated datasets or isolated task-based experiments rather than interactive conversational settings (Arian et al., 2024; Waaler et al., 2025). Furthermore, most evaluations focus primarily on response accuracy or hallucination reduction, while practical system considerations such as token consumption, response latency, and knowledge-base scalability receive limited empirical analysis. In particular, few studies systematically examine how performance changes as the size of the knowledge base increases, even though real-world systems must operate with continuously expanding informational repositories.

For the context of Philippine dental clinics, these limitations have direct implications. While prior research demonstrates that both RAG and prompt engineering can improve LLM responses in domain-specific settings (Thorat et al., 2024), relatively few studies evaluate these techniques within patient-facing conversational workflows, such as appointment booking or service inquiries. Moreover, multilingual conversational environments—where users frequently employ mixed languages such as Taglish, informal expressions, and typographical errors—remain largely unexplored in existing research. These gaps highlight the need for controlled comparative evaluation of contextual optimization strategies in operational chatbot environments.

This study addresses these gaps by systematically comparing prompt engineering and retrieval-augmented generation within a simulated dental clinic customer-service chatbot. Unlike many previous studies, the evaluation focuses on multi-turn conversational workflows, knowledge-base scalability, and operational performance metrics such as response latency and token consumption, providing a more comprehensive analysis of how these techniques perform in realistic service-oriented conversational systems.

Chapter III

METHODOLOGY

Research Design

This study employs an experimental comparative research design to evaluate two optimization techniques for large language model (LLM)-based conversational agents: Prompt Engineering and Retrieval-Augmented Generation (RAG). The design focuses on examining the relative effectiveness of both methods in enhancing the accuracy, robustness, scalability, and efficiency of AI-powered customer service agents for Philippine dental clinics.

Both techniques are implemented under identical conditions using the same underlying LLM backend to ensure that observed differences in performance can be attributed solely to the optimization approach rather than variations in the model itself. Each configuration is tested on identical query sets derived from simulated patient interactions, enabling direct comparison through quantitative and qualitative evaluation metrics. The conversational agent was deployed in a controlled chat-based web environment simulating real-world patient–clinic interactions. The setup replicates typical customer service workflows (e.g., booking appointments, inquiries on services, follow-ups) while ensuring reproducibility and controlled testing conditions.

Variables of the Study

To systematically evaluate the comparative performance of Prompt Engineering and Retrieval-Augmented Generation (RAG), this study identifies and categorizes the independent and dependent variables that define the experimental framework. Clearly specifying these variables ensures structured measurement and isolates the impact of contextual optimization strategy on system performance.

Independent Variable

The primary independent variable in this study is the contextual optimization technique applied to the LLM-based conversational agent:

- **Prompt Engineering**
- **Retrieval-Augmented Generation (RAG)**

Secondary Independent Variables

To examine scalability and robustness under varying operational conditions, the following secondary factors are included:

- **Knowledge-base size**
    - Small (30 entries)
    - Medium (100 entries)
    - Large (300 entries)
- **Input condition**
    - **Clean user queries**

  - **Noisy multilingual inputs, including Taglish, typographical errors, informal abbreviations, and incomplete phrasing**
- **Conversation structure**
  - **Single-turn scenarios**
  - **Multi-turn conversational workflows**

Dependent Variables

System performance will be evaluated using the following measurable outputs:

- Response accuracy
- Hallucination rate
- Context retention score
- Robustness to noisy input
- Latency (response time in milliseconds)
- Token usage and computational cost
- Scalability

Dataset and Knowledge Base

Data Sources

Publicly accessible materials from dental clinic websites, Facebook pages, and online FAQs were collected to construct the foundational knowledge base. These materials include operational details, service descriptions, pricing information, clinic policies, and appointment-related procedures. To evaluate conversational performance, simulated patient interaction scenarios were independently constructed to represent common use cases such as appointment booking, rescheduling, cancellations, and service inquiries.

Knowledge Base Composition

In this study, a *knowledge-base entry* is defined as a single atomic informational unit stored as an independent record (e.g., one FAQ question–answer pair, one service record, or one policy rule). The knowledge base contains structured entries categorized into:

- Frequently Asked Questions (FAQs)
- Service and pricing information
- Clinic policies and operational hours
- Static scheduling information and appointment policies
- Treatment and procedure descriptions
- Pre-procedure preparation guidelines
- Post-procedure care instructions
- Insurance and payment policies
- Dentist profile information (e.g., specialization, qualifications)
- General clinic information and facilities

To evaluate scalability, the knowledge base is configured at three sizes:

- **Small:** 30 entries
- **Medium:** 100 entries
- **Large:** 300 entries

The medium and large configurations are constructed by expanding distinct service offerings, pricing variations, policy rules, and scheduling constraints to simulate realistic growth in informational volume while avoiding duplicated or paraphrased entries.

Conversational Scenario Design

Separate from the static knowledge base, conversational scenarios were constructed to simulate real-world patient–clinic interactions. Each scenario represents a structured patient inquiry or workflow, such as general clinic inquiries, service and pricing questions, appointment booking, rescheduling, cancellation, or ambiguous follow-up messages.

Scenarios were categorized according to input condition and conversation structure. Input conditions included clean queries and noisy multilingual inputs containing Taglish expressions, typographical errors, informal phrasing, abbreviations, and incomplete messages. Conversation structures included both single-turn and multi-turn scenarios.

For multi-turn scenarios, only the patient/user messages were predefined in the scenario dataset. Assistant responses were not manually scripted. During execution, each chatbot configuration generated a response for every user turn. The generated assistant response was then appended to the conversation history before the next user turn was processed. This allowed the experiment to evaluate not only final-answer accuracy but also how each configuration maintained factual consistency and contextual continuity throughout the conversation.

Data Preparation

Knowledge-base entries are cleaned, structured, and standardized to ensure consistency and accuracy. Text is formatted uniformly and duplicates are removed. The curated information is stored in a PostgreSQL database and exported into structured JSON format for experimental use.

Ethical Considerations

All data used in the study are publicly available and anonymized. No personally identifiable or sensitive health information is included. The dataset is used solely for academic research purposes.

System Architecture and Implementation

Two chatbot configurations were implemented for comparative evaluation—one using Prompt Engineering and another using Retrieval-Augmented Generation (RAG)—while maintaining a common architecture for fairness of comparison.

Both chatbot configurations were implemented within a full-stack Next.js application using React-based frontend components and server-side API routes for backend processing and experiment execution. The application was developed using TypeScript to improve type safety and maintainability. Data was managed through PostgreSQL and accessed using Prisma ORM, enabling structured storage of experimental results, scenario metadata, and system logs.

The generative model used for all experimental runs is GPT-4o mini accessed through the OpenAI API.

Prompt Engineering Configuration

In the Prompt Engineering setup, the system constructs the model prompt by injecting: (1) structured system instructions, (2) the last **N** conversation turns, and (3) the full knowledge base content formatted as structured text entries. The resulting prompt is then passed to the LLM to generate a response.

RAG Configuration

In the RAG setup, all knowledge-base entries are embedded *prior to evaluation runs* using **text-embedding-3-large**. During inference, the system embeds the user query and retrieves the **top-k** most semantically relevant entries (**k = 5**) from the vector index using cosine similarity. Only these retrieved entries, together with system instructions and the last **N** conversation turns, are appended to the LLM prompt before response generation.

Shared System Pipeline

The shared system pipeline operates as follows:

1. The user sends a message through the web-based chat interface.
2. The backend determines the active configuration (Prompt Engineering or RAG).
3. The system constructs the prompt according to the selected configuration.
4. The LLM generates a reply, which is sent back to the user through the frontend interface.

5. All interactions—including query text, generated response, response time, and token usage—are logged for evaluation and analysis.

This architecture ensures a controlled environment for comparative testing by isolating differences in contextual injection strategy (direct prompt-based injection vs retrieval-based grounding) while keeping the generative backbone and deployment environment constant.

Algorithmic Framework for Prompt Engineering and Retrieval-Augmented Generation

This study evaluates two distinct algorithmic strategies for supplying contextual information to a large language model (LLM): Prompt Engineering and Retrieval-Augmented Generation (RAG). While both approaches utilize the same underlying generative model, they differ fundamentally in how domain knowledge is incorporated into the prompt during inference. This section formally describes the algorithmic workflow of each configuration and the key parameters governing system behavior.

Prompt Engineering Algorithm

In the Prompt Engineering configuration, contextual knowledge is directly injected into the model prompt at runtime. The full knowledge base is concatenated with conversational history and system instructions, without semantic retrieval filtering. The algorithm proceeds as follows:

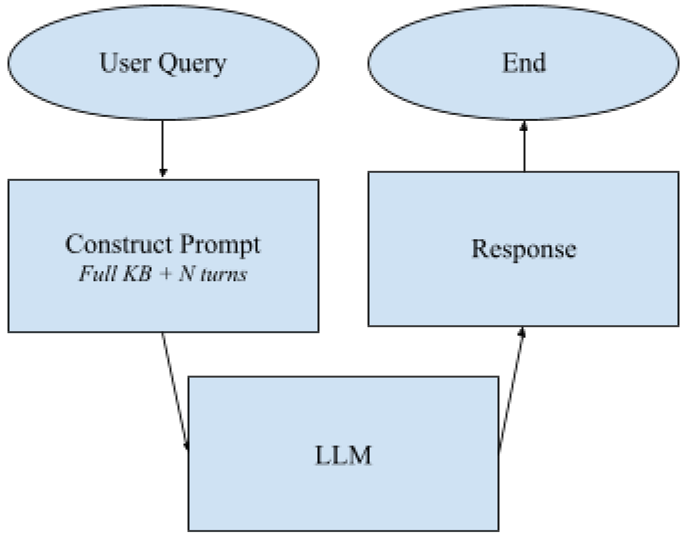

Figure 1. *Prompt Engineering Algorithm Diagram*

Input Acquisition

The system receives a user query along with the existing conversation history.

Context Construction

The last **N** conversational turns are retained to preserve dialogue continuity. The complete knowledge base—formatted as structured atomic entries—is appended to the prompt as static contextual information.

Prompt Assembly

A structured prompt is constructed consisting of:

- System-level instructions defining role and response constraints
- The retained conversational turns
- The full knowledge-base content

Response Generation

The assembled prompt is submitted to the LLM for response generation.

Output Delivery and Logging

The generated response is returned to the user interface. Metadata, including input token count, output token count, response latency, and conversation state, are logged for analysis.

The principal parameters influencing this approach include:

- The number of retained conversational turns (**N**)
- The total size of the knowledge base (30, 100, or 300 entries)
- Prompt formatting and ordering

As the knowledge base grows, the total prompt length increases proportionally. This may lead to higher token consumption, increased latency, and potential degradation in contextual fidelity due to attention dilution as informational volume expands.

Retrieval-Augmented Generation (RAG) Algorithm

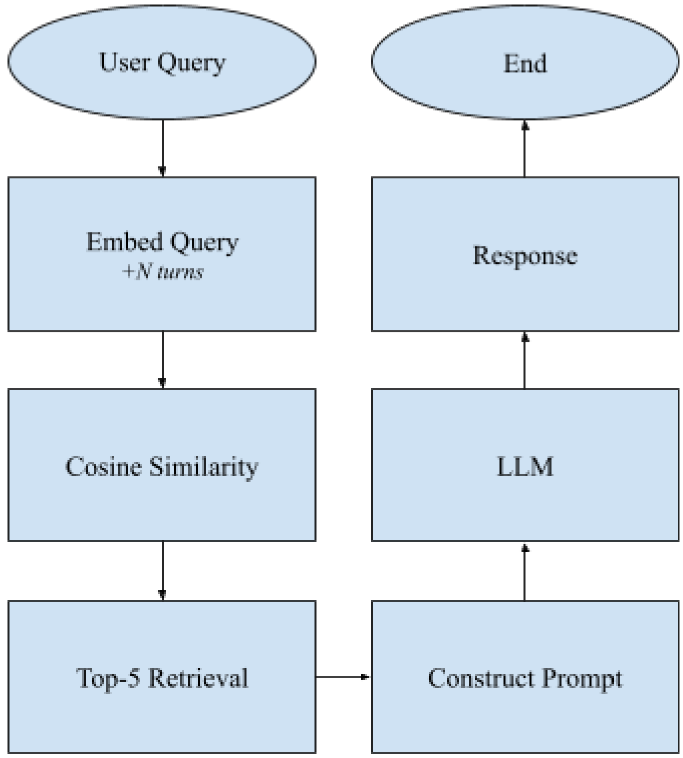

Figure 2. *Retrieval-Augmented Generation (RAG) Algorithm Diagram*

In the RAG configuration, contextual information is dynamically retrieved from the knowledge base at inference time rather than being fully injected into the prompt.

Offline Embedding Generation

Prior to experimental runs, all knowledge-base entries are converted into vector representations using the **text-embeddings-3-large** embedding model. These embeddings are stored in a vector index to enable efficient similarity-based retrieval.

Input Encoding

At inference time, the incoming user query is transformed into a vector representation using the same embedding model.

Semantic Retrieval

Cosine similarity is computed between the query embedding and all stored knowledge-base embeddings. The top-k most semantically similar entries (**k = 5**) are retrieved from the index.

Context Assembly

The retrieved entries are combined with:

- System-level instructions
- The last **N** conversational turns

to construct the final LLM prompt.

Response Generation

The LLM generates a response conditioned only on the retrieved contextual entries and conversation history.

Output Delivery and Logging

The generated response, input/output token usage, and latency metrics are recorded for analysis.

Key parameters governing the RAG configuration include:

- Embedding model (text-embeddings-3-large)
- Similarity metric (cosine similarity)

- Retrieval depth (k = 5)
- Number of retained conversation turns (N)

Unlike Prompt Engineering, RAG prevents prompt length from increasing proportionally with knowledge-base size, as only the top-k relevant entries are injected into the prompt. However, it introduces additional computational overhead from embedding and similarity search operations.

Parameter-Driven Trade-offs and Research Relevance

The two contextualization strategies introduce contrasting trade-offs under increasing knowledge-base size.

Prompt Engineering scales linearly with informational volume, leading to rising token usage and potential latency increases as contextual content grows. In contrast, RAG decouples prompt length from total knowledge-base size by retrieving only a limited subset of relevant entries, thereby potentially improving scalability and computational efficiency.

These trade-offs form the basis of the research problem addressed in this study. As conversational systems for healthcare customer service expand in informational scope, developers must determine whether direct prompt-based context injection remains effective, or whether retrieval-based grounding provides measurable advantages in accuracy, robustness, and efficiency.

By systematically varying knowledge-base size and input characteristics, this study evaluates how each algorithm balances performance and resource efficiency under realistic multilingual customer-service conditions.

Experimental Procedure

The experimental procedure was designed to systematically compare the performance of both chatbot configurations under controlled conversational conditions. The experiment was divided into four phases.

Phase 1 – Scenario Definition

In this phase, standardized test scenarios were constructed to simulate common patient interactions within dental clinics. These scenarios covered five major categories of queries: general inquiries such as clinic hours and location; service-related questions about treatments or pricing; appointment scheduling and modification requests; ambiguous or incomplete inputs (for example, short queries such as “How much?”); and noisy inputs that include Taglish expressions, typographical errors, or informal phrasing. Selected scenarios were structured as multi-turn conversations to evaluate contextual continuity. The same fixed set of scenarios was used across all experimental conditions to ensure consistency.

Representative Test Scenarios

Table X presents representative examples of test scenarios used during the evaluation process. These scenarios were designed to reflect common patient–clinic communication patterns observed in dental service inquiries.

Category	Example User Query

General Inquiry	"What time does the clinic open?"
Service Question	"Magkano po ang dental cleaning?"
Appointment Request	"Can I book an appointment for tomorrow afternoon?"
Appointment Modification	"Pwede bang i-reschedule yung appointment ko sa Friday?"
Ambiguous Query	"How much?"
Noisy / Taglish Input	"Magkano cleaning nyo? may slot tom?"

These scenarios represent a subset of the evaluation cases used in the experiment. They were designed to simulate realistic patient–clinic interactions; however, they are not exhaustive and may be expanded with additional variations to further evaluate system performance under different conversational conditions.

Phase 2 – Execution

Each scenario was processed by both chatbot configurations using the same LLM backend and identical model parameters. In the Prompt Engineering configuration, the full knowledge base was injected into the prompt together with recent conversational context. In the Retrieval-Augmented Generation (RAG) configuration, knowledge-base entries were embedded in advance using the **text-embeddings-3-large** model, and the top five most semantically similar entries were retrieved at inference time using cosine similarity.

The generated responses were automatically stored in a database along with corresponding metadata, including latency and token usage, to enable consistent quantitative analysis.

For single-turn scenarios, the chatbot generated one response for the given user query. For multi-turn scenarios, the system processed each user turn sequentially. After each user message, the selected chatbot configuration generated an assistant response, which was logged and appended to the conversation history before the next user message was submitted. This procedure allowed the experiment to evaluate generated responses at each conversational step while preserving the evolving dialogue context.

Phase 3 – Knowledge Base Scaling

To assess scalability, both systems were tested against three knowledge-base sizes: small (30 entries), medium (100 entries), and large (300 entries). This scaling procedure allowed observation of changes in response accuracy, hallucination behavior, latency, and letoken consumption as the amount of available information increased. The same scenario set was executed across all size conditions to preserve experimental fairness.

Phase 4 – Evaluation and Logging

All collected responses were subjected to structured evaluation. Outputs were assessed based on factual accuracy relative to the ground-truth knowledge base, coherence in multi-turn dialogue, and contextual appropriateness. Instances of hallucination—defined as the generation of unsupported or fabricated information not present in the knowledge base or conversation context—were identified and recorded. Quantitative metrics including token usage and response latency were analyzed alongside qualitative scoring to support comparative performance evaluation.

Control Variables

To ensure experimental fairness and isolate the effects of the contextual optimization technique, several variables were held constant across all experimental conditions. By controlling these factors, any observed differences in performance can be more reliably attributed to the contextualization strategy—Prompt Engineering or Retrieval-Augmented Generation (RAG)—rather than to external, architectural, or model-related variations.

The following control variables were maintained:

Underlying Language Model:

Both configurations utilized the same LLM backend and model version for all experimental runs to eliminate differences arising from model architecture or training data.

Model Hyperparameters:

Generation parameters, including temperature, maximum token limit, and other decoding settings, were kept identical across all test runs.

Embedding Model and Retrieval Parameters:

For the RAG configuration, the text-embeddings-3-large embedding model was consistently used across all knowledge-base sizes. The similarity metric (cosine similarity) and retrieval depth (k = 5) were fixed for all experiments to prevent variability in retrieval behavior.

Query Set:

Each configuration was evaluated using the exact same standardized test scenarios and input variations to ensure direct comparability.

Conversation Turn Limit:

The number of retained conversational turns (N) included in prompt construction was fixed across both configurations to maintain consistent contextual depth.

Knowledge-Base Structure:

Knowledge-base entries were defined as atomic informational units and maintained consistent formatting and structure across all size conditions.

Hardware and Deployment Environment:

All experiments were executed within the same server and application environment to prevent discrepancies in latency caused by infrastructure variability.

Evaluation Rubric:

The same evaluation criteria and scoring procedures were applied uniformly to all generated responses to ensure consistency in qualitative assessment.

By explicitly controlling these variables, the study minimizes confounding effects and strengthens internal validity. This ensures that the independent variable under investigation—the contextualization strategy—remains the primary determinant of observed system performance.

Model Testing and Evaluation Metrics

As this study evaluates pre-trained large language models rather than developing new models from scratch, the testing process emphasizes structured comparative evaluation instead of parameter optimization.

System performance is assessed using both quantitative and rubric-based qualitative metrics. All metrics are computed consistently across configurations and knowledge-base sizes.

Quantitative and qualitative evaluation metrics are summarized as follows:

Metric	Description	Method of Measurement
Accuracy	Degree to which generated responses correctly reflect ground-truth knowledge base information and conversational constraints	Rubric-based scoring using a three-level correctness scale.

Hallucination Rate	Proportion of responses containing unsupported or fabricated information not present in the knowledge base or conversation context.	Manual classification using predefined hallucination criteria; percentage of hallucinated responses computed.
Context Retention	Proportion of responses in later conversational turns that remain consistent with constraints established earlier in the dialogue.	Evaluator scoring of contextual consistency across multi-turn scenarios.
Robustness	Difference in system accuracy between clean and noisy input conditions.	Accuracy_clean − Accuracy_noisy.
Latency	Time in milliseconds between user input and model response generation.	Automatically logged during testing.
Token Cost	Total number of prompt and completion tokens consumed per interaction.	API usage logs aggregated per condition.
Scalability	Performance variation as knowledge base size increases.	Comparative analysis across small (30), medium (100), and large (300) KB conditions.

Operational Definition of Metrics

Accuracy

Accuracy is evaluated using a three-level scoring rubric to capture varying degrees of response correctness. Each generated response is assigned one of the following scores:

2 – Fully Correct: The response provides accurate information that fully addresses the user's query and is consistent with the knowledge base and conversational context.

1 – Partially Correct: The response contains generally relevant information but is incomplete, ambiguous, or omits important details required to fully answer the query.

0 – Incorrect: The response provides incorrect information, fails to address the query, or contradicts the knowledge base or conversation context.

Accuracy scores are computed by summing the assigned scores and normalizing them against the maximum possible score across all evaluated responses.

Accuracy (%) = (Total score obtained / Maximum possible score) × 100

Hallucination Rate

Hallucination Rate (%) = (Number of responses containing unsupported information / Total responses) × 100

Unsupported information refers to claims not present in the knowledge base or established conversational context.

Context Retention

Context retention measures how effectively the chatbot maintains and applies relevant information established in earlier conversational turns during multi-turn scenarios. This includes remembering previously mentioned services, appointment details, scheduling constraints, user intent, or ambiguous references such as pronouns and follow-up questions.

Context retention is evaluated using a three-level rubric-based scoring system:

2 – Fully Retained: The response correctly remembers and applies all relevant information established in earlier turns while maintaining conversational consistency.

1 – Partially Retained: The response demonstrates partial awareness of earlier conversational context but omits, weakens, or misinterprets an important detail.

0 – Not Retained: The response ignores, contradicts, or fails to apply important information established earlier in the conversation.

For single-turn scenarios where no prior conversational context exists, context retention is marked as Not Applicable (N/A).

Robustness

Robustness measures how well each system maintains performance when processing noisy or multilingual inputs. It is calculated as the difference in response accuracy between clean input conditions and noisy input conditions (e.g., Taglish expressions, typographical errors, or incomplete queries).

$$\textbf{Robustness Drop} = \textbf{Accuracy\_clean} - \textbf{Accuracy\_noisy}$$

A smaller robustness drop indicates greater resilience to input noise.

Latency and Token Cost

Latency and token consumption are automatically recorded through system logging and averaged across experimental runs.

Scalability

Scalability is evaluated by analyzing how accuracy, hallucination rate, latency, and token usage change as knowledge-base size increases from 30 to 300 entries.

Human Evaluation Procedure

Qualitative assessments are conducted by two independent evaluators using a predefined scoring rubric. Responses are evaluated for factual accuracy, contextual consistency, and hallucination presence. In cases of disagreement, evaluators review responses jointly until consensus is reached. This process ensures consistent application of evaluation standards across all experimental conditions.

Data Analysis

Data analysis combines descriptive statistical computation with structured qualitative interpretation. For each experimental condition (technique × knowledge-base size × input type), quantitative metrics including accuracy, hallucination rate, latency, and token consumption will be computed as mean values across all scenario runs.

Comparative analysis will be performed to identify performance differences between Prompt Engineering and Retrieval-Augmented Generation (RAG). Where appropriate, statistical tests such as independent sample t-tests will be applied to determine whether observed differences in key metrics (e.g., accuracy or latency) are statistically significant. All statistical tests will be conducted under consistent confidence thresholds.

Performance trends across knowledge-base sizes (30, 100, 300 entries) will be analyzed to assess scalability effects. Differences between clean and noisy input conditions will be examined to measure robustness.

Results will be summarized using descriptive tables and bar charts to visualize differences in performance metrics across configurations.

In addition to quantitative analysis, qualitative examination will be conducted on selected response samples. Hallucination cases and context-retention failures will be reviewed to identify recurring error patterns and contextual breakdown points. This mixed-method approach supports both numerical comparison and interpretive insight into system behavior.

Summary

This methodology establishes a controlled experimental framework for comparing Prompt Engineering and Retrieval-Augmented Generation (RAG) under identical generative and deployment conditions. By systematically varying knowledge-base size and input characteristics while holding model parameters constant, the study isolates the impact of contextualization strategy on system performance.

Through structured quantitative metrics and rubric-based qualitative evaluation, the framework measures differences in accuracy, hallucination behavior, context retention, latency, and token consumption. The design emphasizes scalability, multilingual robustness, and computational efficiency, aligning experimental findings with practical considerations for deploying AI-powered conversational systems in Philippine dental clinic environments.

REFERENCES

[1] P. Lewis, E. Perez, A. Piktus, F. Petroni, V. Karpukhin, N. Goyal, H. Küttler, M. Lewis, W. Yih, T. Rocktäschel, S. Riedel, and D. Kiela. 2020. Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks (RAG). NeurIPS. https://proceedings.neurips.cc/paper/2020/file/6b493230205f780e1bc26945df7481e5-Paper.pdf

[2] S. Gupta, R. Babu, and P. Tiwari. 2024. A Comprehensive Survey of Retrieval-Augmented Generation (RAG): Evolution, Current Landscape and Future Directions. arXiv preprint. https://arxiv.org/pdf/2410.12837

[3] W. Fan, L. Kong, and S. Sun. 2024. A Survey on RAG Meeting LLMs: Towards Retrieval-Augmented Large Language Models. ACM Computing Surveys. https://dl.acm.org/doi/10.1145/3637528.3671470

[4] V. Thorat, H. Patil, and V. Bhuse. 2024. Role of Artificial Intelligence (AI) in Patient Education and Communication in Dentistry. J. Dent. Res. https://pmc.ncbi.nlm.nih.gov/articles/PMC11155216/

[5] A. Arian, S. Heston, and M. Bhuse. 2024. Dental Loop Chatbot: A Prototype Large Language Model Framework for Dentistry. J. AI Dent. https://www.mdpi.com/2674-113X/3/4/29

[6] P. N. Waaler, T. H. Haug, and K. R. Olsen. 2025. Prompt Engineering an Informational Chatbot for Education on Mental Health Using a Multiagent Approach. JMIR AI. https://pmc.ncbi.nlm.nih.gov/articles/PMC11982747/

[7] A. Singh and A. Namin. 2025. A Survey on Chatbots and Large Language Models: Testing and Evaluation Techniques. J. Syst. Softw. https://www.sciencedirect.com/science/article/pii/S2949719125000044

[8] Z. Yang, L. Xu, and F. Sun. 2025. RAGVA: Engineering Retrieval Augmented Generation – An Experience Report. J. Syst. Softw. https://www.sciencedirect.com/science/article/pii/S0164121225001049

[9] S. Shin, C. Lee, and H. Kim. 2024. Charting the Evolution of Artificial Intelligence Mental Health Chatbots from Rule-Based Systems to Large Language Models: A Systematic Review. Frontiers in Digital Health. https://pubmed.ncbi.nlm.nih.gov/40948070/

[10] K. D. Singh, R. K. Sharma, and P. Kumar. 2022. Assisting Personalized Healthcare of Elderly People: Developing a Rule-Based Virtual Caregiver System Using Mobile Chatbot. Journal of Ambient Intelligence and Humanized Computing. https://pubmed.ncbi.nlm.nih.gov/35632238/

[11] A. M. Holst and L. F. Iala. 2024. How the Choice of LLM and Prompt Engineering Affects Chatbot Effectiveness. Electronics (MDPI) 14(5): 888. https://www.mdpi.com/2079-9292/14/5/888

[12] J. Wang, et al. 2025. Balancing Accuracy and User Satisfaction: The Role of Prompt Engineering in AI-Driven Healthcare Solutions. https://pmc.ncbi.nlm.nih.gov/articles/PMC11865202/

[13] H. Patil, S. Heston, and M. Bhuse. 2024. Prompt Engineering in Healthcare. Electronics (MDPI). https://www.mdpi.com/2079-9292/13/15/2961

[14] J. Steybe, et al. 2025. Evaluation of a Context-Aware Chatbot Using Retrieval-Augmented Generation for Medication-Related Osteonecrosis of the Jaw (MRONJ) Clinical Questions. https://www.sciencedirect.com/science/article/pii/S101051822400341X

[15] Y. Qi, et al. 2024. RAG-Optimized Tibetan Tourism LLMs: Enhancing Accuracy and Personalization. arXiv preprint. https://arxiv.org/abs/2408.12003

[16] V. Wüst and P. Bremser. 2025. Artificial Intelligence in Tourism Through Chatbot Support in the Booking Process. https://www.mdpi.com/2673-5768/6/1/36

[17] S. Siriwardhana, et al. 2023. Improving the Domain Adaptation of Retrieval Augmented Generation (RAG) Models for Open Domain Question Answering. https://www.researchgate.net/publication/367403483_Improving_the_Domain_Adaptation_of_Retrieval_Augmented_Generation_RAG_Models_for_Open_Domain_Question_Answering

[18] J. Xu, et al. 2024. SimRAG: Self-Improving Retrieval-Augmented Generation for Adapting Large Language Models to Specialized Domains. arXiv preprint. https://arxiv.org/abs/2410.17952

[19] T. Mertelj and BEKO Solutions. 2025. Retrieval-Augmented Generation: Empowering AI with Real-Time Context & Intelligence (White Paper). https://beko-solutions.si/wp-content/uploads/2025/07/BEKO-Insights_RAG-Systems-Whitepaper_Final.pdf

[20] P. Gao, H. Zhang, Z. Yang, and Z. Chen. 2023. Retrieval-Augmented Generation for Large Language Models: A Survey. *arXiv preprint*. https://arxiv.org/abs/2312.10997

[21] R. Pradhan, "RAG vs. Fine-Tuning vs. Prompt Engineering: A Comparative Analysis for Optimizing AI Models," *International Journal of Computer Trends and Emerging Computing Engineering*, vol. 8, no. 5, Oct. 2025, doi: 10.15680/IJCTECE.2025.0805004.